DIVYANSHU BHATIWAL

B.Tech CSE (AI & ML) · SRM Institute of Science and Technology, Delhi-NCR · Batch 2027
+91 88520 99179 · db7694@srmist.edu.in · linkedin.com/in/divyanshu-bhatiwal · github.com/divyanshubhatiwal

Summary

Backend-focused CSE (AI & ML) student who ships production-shaped systems end to end: REST services on Node.js and Python, relational and NoSQL data layers, authentication and encryption, containerised deploys and CI. Comfortable translating a design into working code with minimal guidance, reasoning about concurrency, failure modes and idempotency, and building agentic LLM workflows rather than just calling an API. Strong DSA foundation in C/C++ and Java.

Technical Skills

- **Languages:** Java, JavaScript, TypeScript, Python, C, C++, SQL, PHP
- **Backend & APIs:** Node.js, Express 5, FastAPI, Flask, REST API design, JWT auth with refresh-token rotation, TOTP 2FA, WebSocket streams, rate limiting and throttling middleware, Celery background workers
- **Databases:** MySQL, PostgreSQL, MongoDB, Redis, Elasticsearch; schema design, indexing, query tuning, migrations, normalisation and referential integrity
- **CS Fundamentals:** Data Structures & Algorithms, Operating Systems (Linux, Windows), DBMS, OOP, Compiler Design, software design and system decomposition
- **AI & Agentic:** Agentic loops (goal manager, planner, critic, episodic memory), LLM integration with deterministic fallbacks, Gemini API, Cursor and GitHub Copilot in daily workflow, Scikit-learn, TensorFlow, NLP
- **Cloud & Tooling:** Docker, GitHub Actions CI/CD, Nginx, AWS, Git/GitHub/GitLab, Jest and pytest suites, Agile/Scrum, technical documentation

Projects

Venture DAO — Autonomous Trading & Governance Platform 2026
Node.js, Express 5, MongoDB, React 19, Docker • github.com/divyanshubhatiwal/venture-dao
- Built the server-side execution engine as an explicit state machine with idempotent order dispatch, so a retried or duplicated request can never open a second position — the core correctness problem in any money-moving service.
- Designed a mandatory risk preflight layer (drawdown ceilings, daily loss limits, position caps, stop-loss enforcement) that no calling path can bypass, alongside an agent loop with goal manager, decision, critic and episodic memory.
- Hardened credentials with AES-256-GCM encryption at rest, scrypt hashing, TOTP second factor and HMAC-signed outbound calls; covered engine, auth, vault and venue adapters with 20+ Jest suites before deploying via Docker.

RideCompare — Multi-Provider Fare Aggregation Service 2026
FastAPI, PostgreSQL 16, Redis 7, Celery, Flutter, Nginx • github.com/divyanshubhatiwal/ridecompare
- Fanned out concurrent fare and ETA lookups across four ride-hailing providers behind a common adapter interface, normalising inconsistent third-party responses into one comparable schema with Redis caching on the hot path.
- Layered JWT access/refresh rotation, Celery workers for price alerts and token cleanup, versioned schema migrations, and a GitHub Actions to Docker deploy pipeline behind an Nginx reverse proxy.

SafeSignal AI — NLP Pharmacovigilance Data Platform 2026
FastAPI, PostgreSQL, Redis, Elasticsearch, React, pytest
- Built an ingestion pipeline that parses heterogeneous adverse-event reports, normalises them into a relational Postgres schema, and applies ROR/PRR disproportionality statistics to surface safety signals.
- Served the analytics layer through FastAPI with Redis caching and Elasticsearch-backed search, covered by a full pytest suite so data-quality regressions are caught before they reach downstream output.

FlowBoard — AI-Powered Project Management App Jan 2026 – Present
React 18, TypeScript, Supabase (PostgreSQL, row-level security) • Live
- Shipped 15+ views on a real-time Postgres backend with row-level security and live sync; integrated an LLM for natural-language task creation with a rule-based fallback for when the model is unavailable.

T20 Score Predictor — End-to-End ML Pipeline Jan 2025 – Present
Python, Scikit-learn, Pandas, Streamlit • Live deployment
- Engineered in-match state features from raw ball-by-ball data, benchmarked Linear Regression against Random Forest, tuned the winner and deployed it as a live public app — full lifecycle owned solo.

Education

B.Tech, Computer Science & Engineering (AI & ML) 2023 – 2027
SRM Institute of Science and Technology, Delhi-NCR Campus, Ghaziabad CGPA: **8.83** / 10
Relevant coursework: Data Structures & Algorithms, Operating Systems, DBMS, OOP, Compiler Design, Software Engineering & Project Management, Data Mining, NLP.
Class XII (CBSE, 2022) — **88.00%** Class X (CBSE, 2020) — **78.66%**

Certifications

Java and Python — HackerRank (2024) · Introduction to Artificial Intelligence — IBM / Coursera (2025) · Infosys Springboard (2025): Data Science; Data Analysis with Python & Pandas; Data Mining; NLP